

Meerkat Sentry

Every roof keeps watch.

HONORABLE MENTION · TAIWAN DEFENSE TECH HACKATHON 2026

TEAM 08 · CHALLENGE 04 · 2–4 OCT 2026

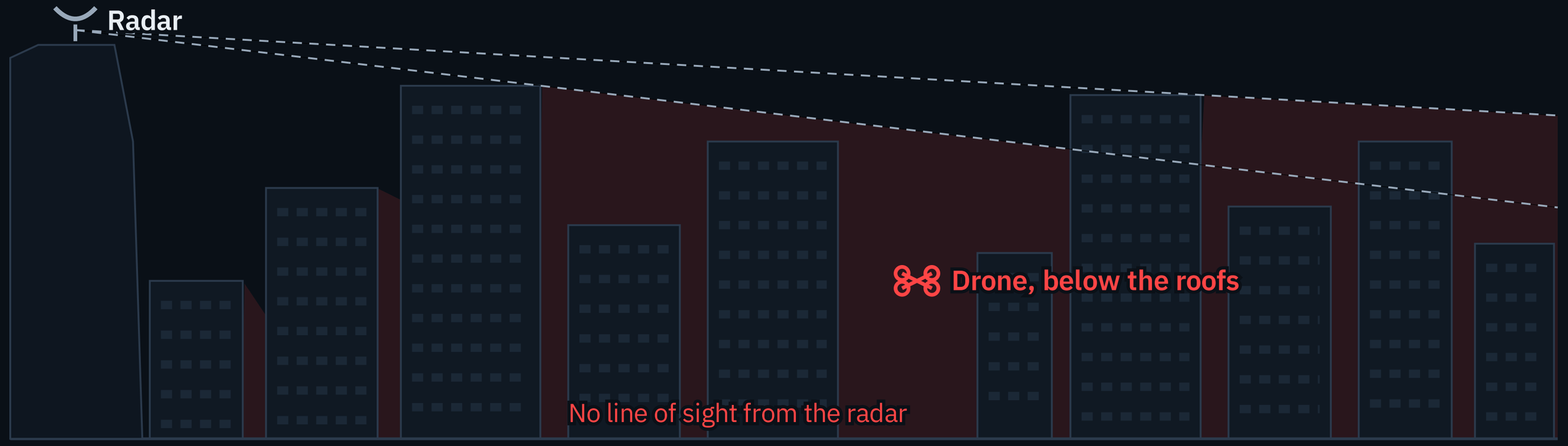
NATIONAL TAIWAN UNIVERSITY, TAIPEI

THE PROBLEM

In a city, small drones fly below the roofs, where radar cannot see

Small drones already disrupt cities: Taoyuan airport stopped its runway (2023, 2026), European airports closed for hours (2025), drones flew over Kinmen outposts (2022).

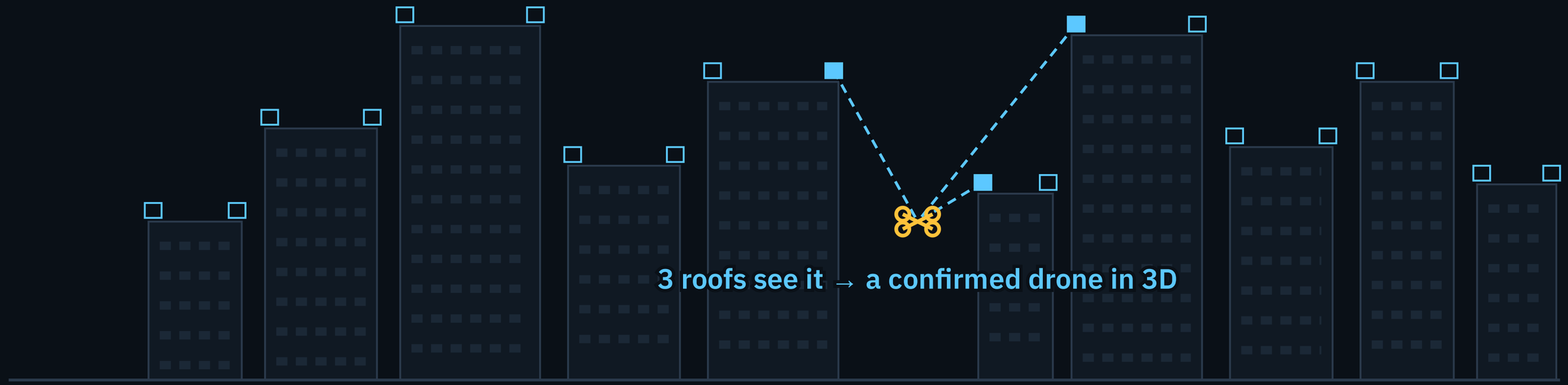
Lesson from Ukraine: cheap drones hit power stations first, then the crews who come to repair them.



Sources: Focus Taiwan (2023), Taipei Times (2026), Kyiv Independent & Wikipedia (2025), Jamestown Foundation (2022).

HOW IT WORKS

Cameras on many roofs look along the streets



1 · ROOF BOX

2–6 cheap cameras per roof. Video stays in the box; only detections leave it.

2 · NETWORK

2 roofs give a 3D position, 3 confirm a drone. Birds, balls and drones that broadcast Remote ID are filtered.

3 · CITY CONTROL ROOM

One alert with the reason: speed, heading, time to target. A person accepts, rejects or changes.

4 · WARNING

The approved alert goes to the authority that warns the public, like an earthquake alert.

REAL · BUILT AT EDTH, 2–4 OCT

Four parts, all built this weekend

ROOF BOX

Raspberry Pi + camera

Finds the drone in its own image at 15–20 fps.

→

CONTROLLER

Laptop

Combines the views into one 3D track, 23 ms after the camera.

→

DASHBOARD

Browser

Rules, alert with the reason, operator decision, log.

→

WARNING

Phone

Demo: Telegram. Product: the alert goes to the warning authority.



OUR LAB · 3 CAMERAS MARK THE SAME DRONE · 20 OF 20 INDOOR FLIGHTS DETECTED

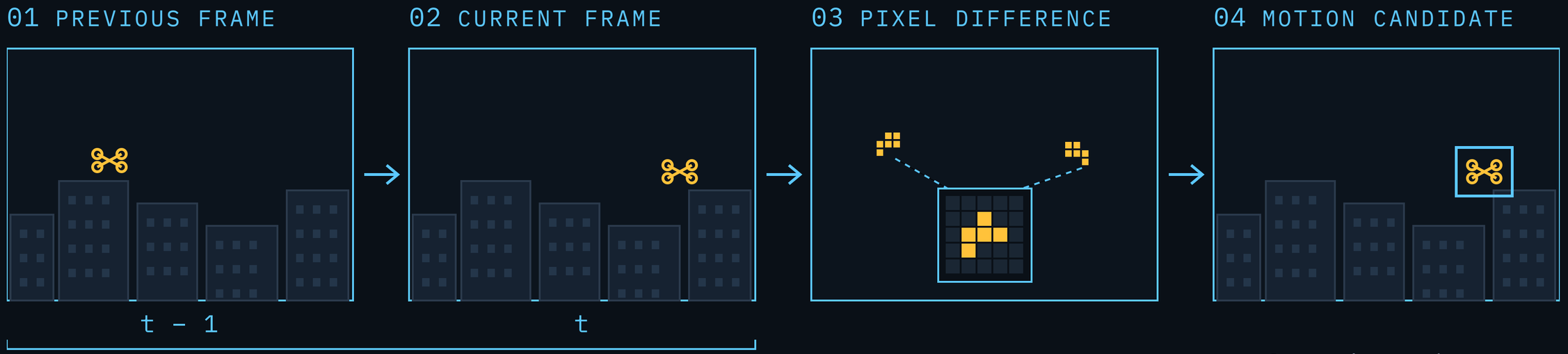
OUTDOOR TEST · 4 OCT · ONE STATION + IMAGE CHECK · OPERATOR CONFIRMED · PHONE ALERTED

Off-the-shelf parts. No core part made in the PRC. Outdoor fusion of 3 roofs: not yet tested.

HOW DETECTION WORKS

Pixel changes reveal moving targets

Each fixed camera compares consecutive frames.



Pixel change detects motion.

Multiple camera views help confirm the target and estimate its 3D position.

REAL · OUTDOOR TEST · 4 OCT 2026

Demo: one drone, from camera to phone

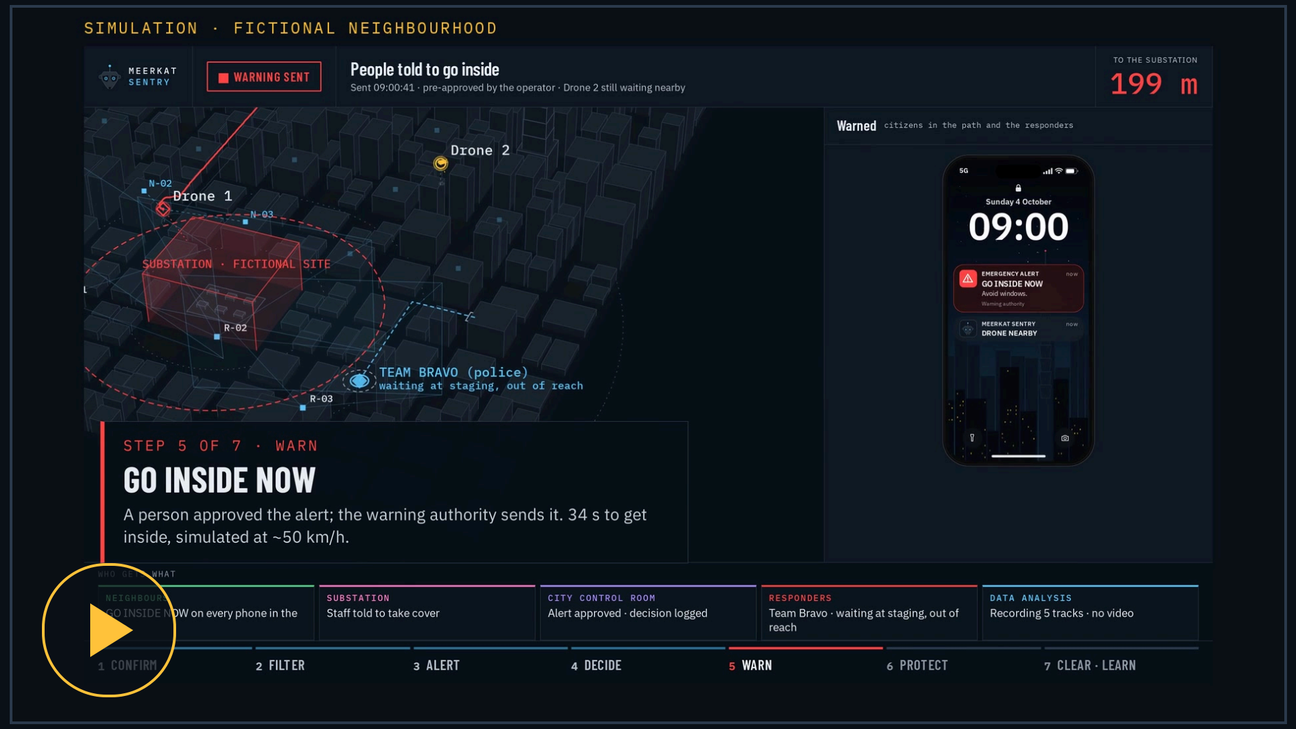


▶ VIDEO · 33 S · IN THE PPTX

1. The drone shows up in a camera image
2. An image check scores it (one station outdoors)
3. The dashboard asks a person: drone or not?
4. The operator confirms; the phone gets the alert

SIMULATION · FICTIONAL NEIGHBOURHOOD

The same software, for a city neighbourhood



▶ VIDEO · 85 S · IN THE PPTX

24 roofs
about 300 m apart (simulation)

1.4 km
drone confirmed this far from the substation

34 s
to get inside if the warning is approved early; 26 s if late (drone at ~50 km/h)

Responders safe
warned in time; the second drone hit an empty gate

STAKEHOLDERS · ONE NETWORK

Residents host, the city runs it, the authority warns

RESIDENTS

Host a roof box

free; they get warnings for their block

roof + power

heads-up

MEERKAT SENTRY

The network

3 roofs confirm a drone; video stays in the box



one alert + reason

CITY CONTROL ROOM

Runs the network

a person checks every alert

approved alert

WARNING AUTHORITY

Warns the public

NCDR or Ministry of National Defense

MILITARY & RADAR

Get the picture

below the radar, as a complement

tracks

Video never leaves the box. Data is never sold. A person always approves a public warning.

BUSINESS MODEL · DUAL USE

One network, two uses, two buyers

	EVERYDAY	CRISIS
USE	Find illegal drone flights; residents see drones near home	Warn people in the drone's path
BUYER	Residents: optional own box, at cost	City / civil defence: base network, control room, service (target buyer)

WHY A CITY PAYS	Taipei already paid for ~13,000 police cameras; Taiwan planned 10,200 low-cost air sensors; Ukraine runs ~10,000 acoustic sensors.
DEMAND EXISTS	Airports and prisons already pay for drone detection.
HOW WE EARN	Boxes at cost. Service per box per month: maintenance, replacement, software, control room. Our cost ≈ $8+ per box/month (assumption); price set in a pilot.

SIZE

~78,000 roof boxes
Taiwan's urban areas, one box per 250 m (assumption)

≈ $2.8 bn
drone detection market by 2030

$345 → ≈ $100
per box today (parts) → target at volume (assumption)

Sources: Taipei Times (2009), Taiwan EPA (2017), Defense One (2024), MarketsandMarkets, Land Management Agency (2022). Assumptions are marked.

RESULT · HONORABLE MENTION

Every roof keeps watch

Honorable Mention at the Taiwan Defense Tech Hackathon 2026. The jury rated originality and scalability highest.

WHAT A PILOT WOULD TEST · range of 3 roofs together · coverage between high-rises · whether a city pays

Jan Barkemeyer linkedin.com/in/janbarkemeyer

Robin Fischer

Edoardo Dominikus

Sean Ching

Camera nodes & fusion: Edoardo Dominikus, Sean Ching · Dashboard, simulation & videos: Jan Barkemeyer · Deck & strategy: Jan Barkemeyer, Robin Fischer · Pitch: Robin Fischer



What the jury said, and what we changed

THE JURY SAID	WE CHANGED
Size of the opportunity and demand rated lowest	Market counted from the bottom; two buyer groups (residents, city)
Emphasise dual use; show civilian and defence buyers	One network, two uses: everyday and crisis; military gets the picture below the radar
Views on urgency were mixed	Incidents from peacetime: Taoyuan, Europe, Kinmen
Originality and scalability rated highest	Kept unchanged: every roof keeps watch

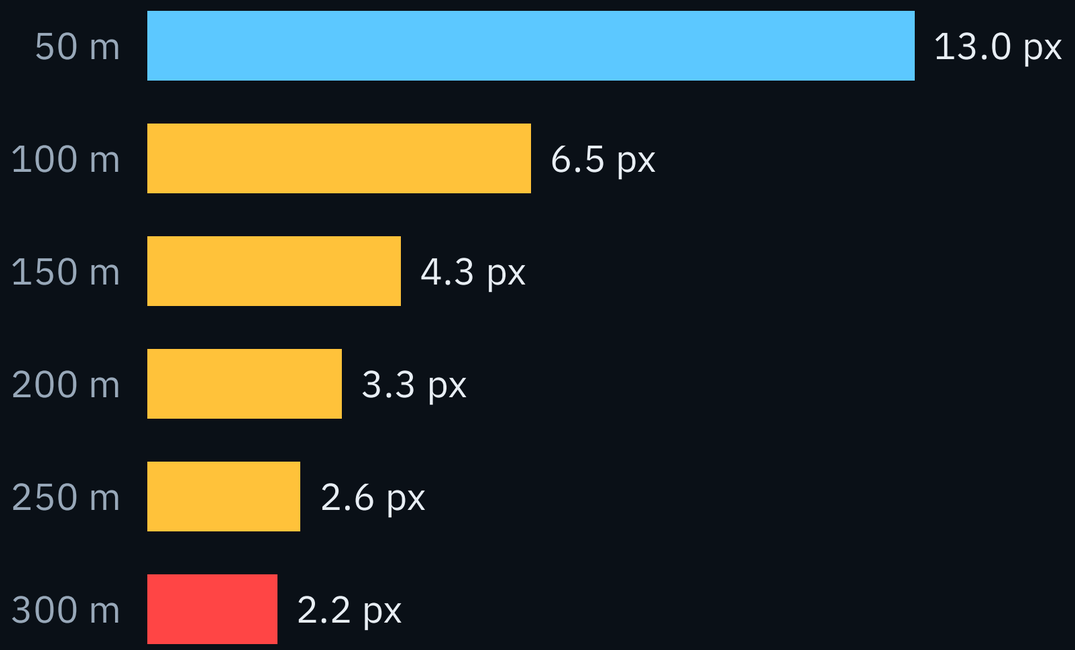
Challenge 04, point by point

CHALLENGE 04 ASKS FOR	WHAT WE BUILT	STATUS
Detect drones in video	3 Pi cameras, controller; 20 of 20 indoor flights	built
Persistent track ID and path	Track IDs and recorded path in the dashboard	built
Alerts: speed, direction, dwell, zone entry	Each alert states its reason in plain words	built
Recommend: track, warn, avoid, allocate	4 recommendations per alert	built
Operator accepts, rejects, modifies	Buttons and a decision log with times	built
Buyer, architecture, scaling	Slides 3, 8 and 9	pitch
No weapon control	Warnings and team allocation only	by design

Field problems and our answers

PROBLEM	OUR ANSWER	STATUS
Birds	Erratic course: filtered before the operator	built
Balls, thrown objects	Fall in an arc: filtered	built
Civil drones with Remote ID	Remote ID received: filtered, unless heading for the site	built
Decoys, one faulty camera	At least 3 independent roofs must agree	built
Night, fog	Thermal camera + microphones: sight, heat, sound	*future*
Between high-rises	Boxes at different heights and on corners; real coverage is a pilot test	not yet tested
Power or internet down	Runs as long as the warning channel runs: small battery, building emergency power, mobile data; box-to-box radio only as last resort	concept
Sun in the lens	Cameras face away from the sun; sun sector masked	designed
Hacking	Boxes send only small detection messages; video never leaves the box	designed
Typhoon	Sealed box, about 10 W, typhoon mount	designed

How far one camera sees a small drone



Pixels across a 35 cm drone with a wide camera (4608 px over 102°): **≈ 650 ÷ distance in m**.

One camera alone needs about 8–10 px: about 80 m. Several roofs that agree can work with 2–3 px: about 200–300 m.

So the boxes sit 250–300 m apart.

Calculated, not yet measured. If range is shorter, boxes stand closer: more boxes, same warning time.

Drone size and pixel thresholds are assumptions.

What the operator never sees

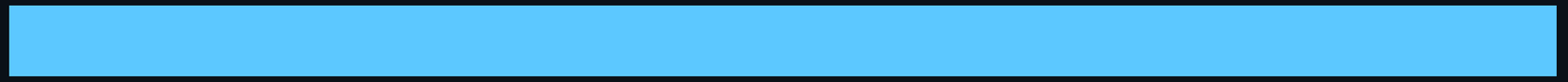
3,752	4	2	1
camera detections	tracks	drones	threat

FILTERED	WHY
60 sensor artefacts	Seen by one camera only
1 ball	Falls in an arc
1 bird	Erratic course
1 Remote ID drone	Broadcasts Remote ID
Never filtered	Anything heading for the protected site

Numbers from our simulated neighbourhood scenario.

Deciding early gives people more time

Operator pre-approves the warning when the drone turns in



34 s

Sent the moment the drone is on course to the site

Operator decides at the last moment



26 s

8 seconds of thinking are 8 seconds less to hide

Time budget: 30 s to take cover + 10 s delivery + 10 s operator. Simulation: drone confirmed 1.4 km out, final dash from 240 m.

Simulation, drone at ~50 km/h.

Who decides which warning

Roof boxes
detect; video stays in the box

→

Network
3 roofs confirm

→

City control room
a person approves

→

Warning authority
NCDR / MND

→

Cell broadcast
“Go inside” to every phone in the area

Taiwan already has the channel: the Public Warning System reaches every phone in an area within seconds and already sends earthquake alerts. The warning authority decides; we deliver a confirmed track with the reason. Telegram is only our demo channel.

What is inside the box

PARTS ONLY (NO INSTALLATION, MOUNT OR DATA LINK)

PART	PRICE	SOURCE
Raspberry Pi 5, 4 GB	$96.49	price Jul 2026
AI HAT+ 13 TOPS (Hailo-8L)	$70	official
2 × Camera Module 3 Wide	$77	PiShop
PoE HAT for Pi 5	$19.99	Waveshare
Housing, window, typhoon mount	$45	estimate
SD card, cables, fan	$20	estimate
PoE injector	$15	estimate
Box S total	**≈ $345**	

Box M = 2 computers, 4 cameras ≈ $650 · Box L = 3 computers, 6 cameras ≈ $950. The PoE HAT is made in China: replace it for government use.

We add to radar, we do not replace it

	PRICE	COVERAGE	EMITS	WARNS PEOPLE
Counter-drone radar	$30k–300k per unit	Blind behind high-rises	Yes: can be located	No
AnyLeaf Meerkat	$1,000–1,500 per site	One site	No	No
Meerkat Sentry	≈ $345 per roof box	A district, grows with every roof	No	Yes: via the warning authority

Cameras need no radio signal: they also see fiber-optic and pre-programmed drones, which RF detectors miss.

“Meerkat” is also the name of AnyLeaf’s single-site detector (anyleaf.org/meerkat). Radar prices: ae-defense.com, unmannedairspace.info.

What every incident teaches the network

QUESTION	WHAT WE LEARN	WHAT CHANGES
Where do they come from?	Approach directions, speeds, time of day	Where the next roofs go
Where do they wait?	Drones hovering before the attack	Patrols check those roofs and parks
When does the second drone strike?	Seconds between impact and second strike	Teams stage out of reach
Where are we blind?	Path parts seen by fewer than 3 roofs	Gaps to close first
How fast were we?	Detection → decision → warning → impact	Operator training

Rules: no video, no faces. Stored in Taiwan. Shared only with the city control room and the warning authority, never sold.

What changed our plan this weekend

CHANGE	WHO OR WHAT CAUSED IT
Coast → city	A drone expert: radar is blind between high-rises
One camera → many roofs	Our outdoor test: against a building the drone was a few pixels
Warn the responders too	A Ukrainian practitioner: a second drone waits for the helpers
2 roofs → 3 roofs to confirm	Our own rule against decoys and faulty cameras

Sources

TOPIC	SOURCE	DOMAIN
Air sensors, Taiwan (10,200 planned)	Taiwan EPA, sensor conference poster (2017)	aqmd.gov
Airport closure, Copenhagen	2025 European drone sightings (2025)	en.wikipedia.org
Airport closure, Munich	Kyiv Independent (2025)	kyivindependent.com
Drone detection market	MarketsandMarkets	marketsandmarkets.com
Illegal drone flights fined (72 cases)	Taipei Times (2026)	taipeitimes.com
Kinmen outposts	Jamestown Foundation (2022)	jamestown.org
Police cameras, Taipei (~13,000)	Taipei Times, 22 Apr 2009 (reprint: TAHR)	tahr.org.tw
Prisons, England & Wales (1,712 sightings)	Boston Globe, UK government data (2026)	bostonglobe.com
Sky Fortress, Ukraine (~10,000 sensors)	Defense One (2024)	defenseone.com
Taoyuan airport, Mar 2023	Focus Taiwan (2023)	focustaiwan.tw
Taoyuan airport, Aug 2026	Taipei Times (2026)	taipeitimes.com
Taoyuan airport detection system	Taiwan News (2023)	taiwannews.com.tw
Urban planning areas (4,869 km²)	Land Management Agency (2022)	nlma.gov.tw
Warning channel	Public Warning System	ncc.gov.tw

Own measurements (fps, latency, indoor flights, outdoor test) from our test logs and video, 3–4 Oct 2026.